

Pointers and the address operator ★ Golden

One college period · 40–45 minutes · Pointers (Sindh XII + C paper)

What you will learn

- Explain & and *
- Declare and initialise a pointer
- Write a program that prints address and value
- Pass addresses to a function to double two numbers

Pointer. A variable that stores the memory address of another variable. Declared with *. The address operator & gives an address; the dereference operator * accesses the value at that address.

```
int x = 10;
int *p;      /* p is a pointer to int */
p = &x;      /* p holds the address of x */
printf("Value = %d\n", *p);  /* 10 */
printf("Address = %u\n", p);
```

```
void double_them(int *a, int *b) {
    *a = *a * 2;
    *b = *b * 2;
}
void main() {
    int x = 4, y = 5;
    double_them(&x, &y);
    printf("%d %d", x, y);  /* 8 10 */
}
```

HOW BIEK ASKS THIS

2025 Paper II: What is a pointer in C? How is it initialised? What is the address operator?

COMMON BOARD MISTAKE

p = &x is wrong for the first assignment if p is int. Use p = &x; then read *p. Mixing them loses marks.

اردو میں سمجھیں

پوائنٹر وہ متغیر ہے جو دوسرے متغیر کا پتہ رکھتا ہے۔ & پتہ دیتا ہے، * اس پتے پر موجود قدر دیتا ہے۔

CHECK YOURSELF

1. If `int x=5; int *p=&x;` what is `*p`?
2. Symbol for address operator?

Answers: 1. 5 · 2. &

MCQs

1. A pointer stores a:
 - a. value only

- b. memory address
- c. file name
- d. keyword

2. &x means:

- a. value of x
- b. address of x
- c. pointer declaration
- d. multiply x

Key: 1=b, 2=b

Short questions (3 marks each)

1. What is a pointer in C? How is it initialised?
2. Explain the address operator with an example.
3. Write a program to display address and value of a variable using a pointer.

Model short answers

1. A variable that holds an address. `int *p; p = &x;` or `int *p = &x;`
2. &x is the address of x. `scanf` uses it; pointers store it.
3. Declare x and `p=&x;` `printf` value *p and address p.

Long questions (board Section C)

1. Explain pointers with a diagram of x and p. Write a function that receives two addresses and doubles both values.